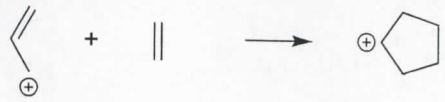
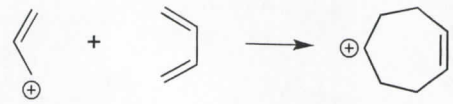
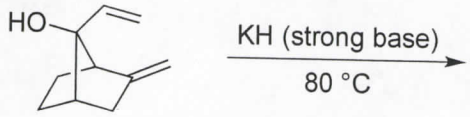
Organic Cume
January 29, 2016

1. According to the Woodward-Hoffmann rules (symmetry of molecular orbitals), predict which of the following cycloaddition reactions are allowed and which are forbidden. Briefly give the reasoning for your answers.

(1) 

(2) 

2. Give the major product of the following reactions. Specify the stereochemistry when applicable.

1) 

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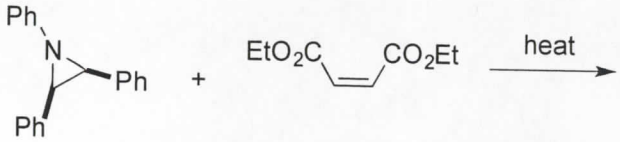
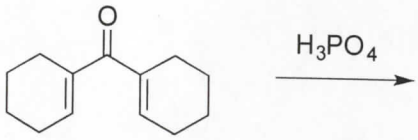
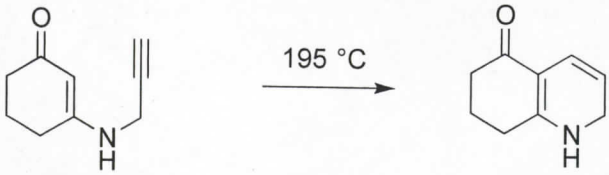
3) 

4) 

3. Write a detailed arrow-pushing mechanism for the following transformation.